49+25·17 = 74·17 / 3·50

Department of Chemistry

Jahangirnagar University

B.Sc. (Hons.) Part II Final Examination 2023

Course Title: Organic Chemistry II; Course no. Chem 230

Full Marks: 70, Time: 4 hrs

[Answer any five questions]

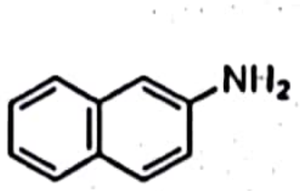
1. ● Treatment of phenanthrene with Br_2 affords a compound with the molecular formula $C_{14}H_{10}Br_2$. Show the reaction. 1

● Explain why- 3+3

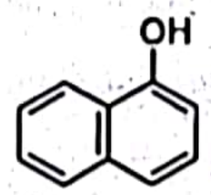
● Nitration of 1-nitronaphthalene takes place in the unsubstituted ring.

● Anthracene is less resistant toward oxidation or reduction than naphthalene.

● Outline all steps in the synthesis of each of the following compounds from naphthalene: 3+2

● 

2- Naphthylamine

● 

α- Naphthol

● Is it possible to convert 2-naphthylamine to naphthalene? Explain your answer. 2

2. ● Propionaldehyde and acetone are isomers of each other: 2+2

● Which test confirms the presence of carbonyl group in each of them? Show the reactions.

● How would you distinguish between them?

● Show the reaction of cyclohexanone with a secondary amine to form an enamine with mechanism. 3

● Why does benzaldehyde undergo Cannizzaro reaction, but not self aldol condensation reaction? 3

● Show the reaction of acetophenone with each of the following reagents- 4

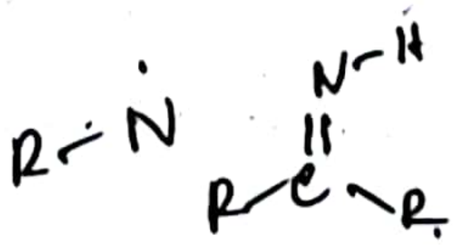
● $LiAlH_4$; ● I_2 + NaOH (both in excess); ● $Ph_3P{=}C(CH_3)_2$; ● NH_2NH_2, KOH, heat.

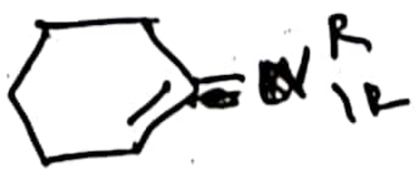
3. (a) What are the carboxylic acids? Give examples. 1

(b) Why is the six carbon atom containing long chain carboxylic acid called caproic acid? 1

(c) How can you chemically distinguish between formic acid and propionic acid? Give necessary chemical equations. 1.5

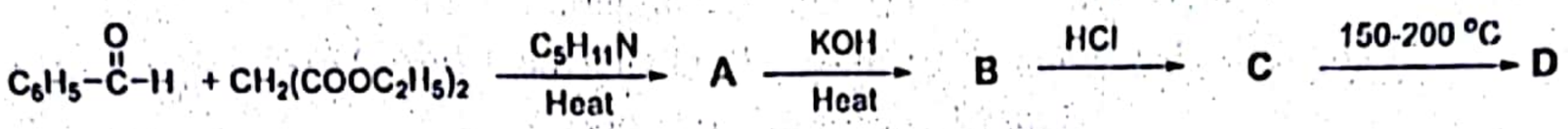
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(d) Draw the structures of the products A-D in the following series of reactions and answer the following questions: 3

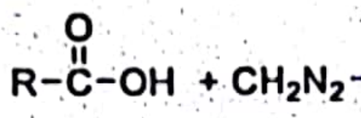


(i) Write down the mechanism of the formation of the product A. 2

(ii) Why is it necessary to use the weak organic base piperidine to carry out the reaction? 1

(iii) Can you use a strong base NaOH or KOH instead of piperidine to carry out the reaction? Give explanation. 2

(e) Write down the product and mechanism of the following reaction. 2.5



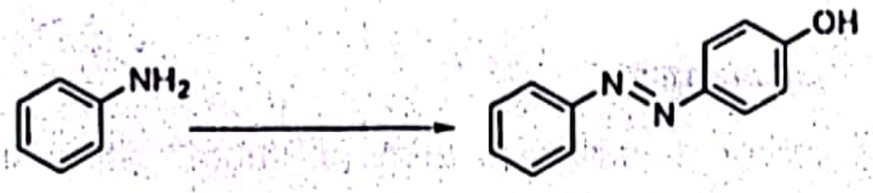
4. Methylamine (CH_3NH_2) is a stronger base than acetonitrile (CH_3CN)-explain why? 2

Outline all the steps in the synthesis of *p*-toluic acid from toluene. 3

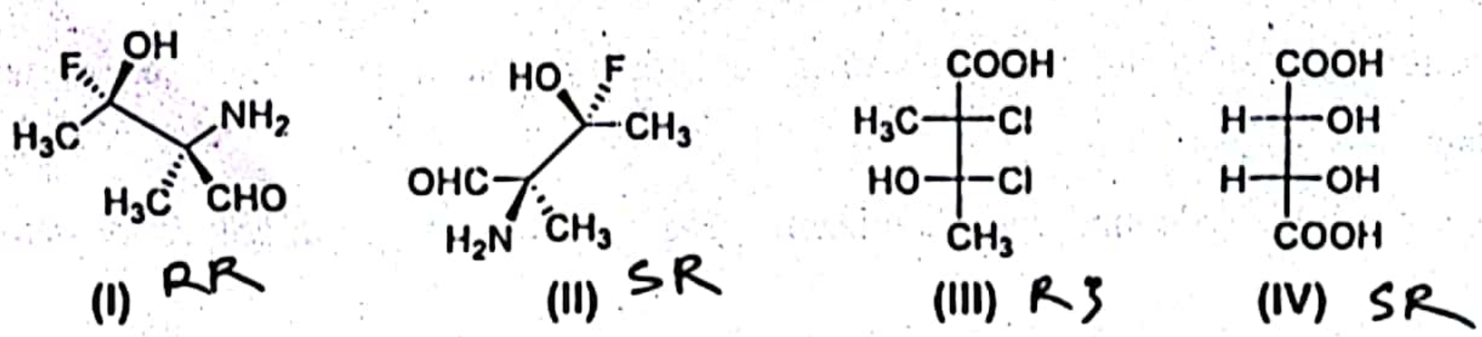
Write down the mechanism of the Hofmann degradation reaction of benzamide. 3

How would you distinguish among 1°, 2° and 3° aromatic amines? 3

Carry out the following conversion: 3



5. Consider the following stereoisomers and answer the related question:



Determine the R/S configuration of each chiral center of the above stereoisomers. 2×4

Which of the above stereoisomers is meso compound? Give reason to support your answer. 3

Write down the difference between enantiomer and diastereoisomer with example. 3

6. (a) What are sulfa drugs? Write down the names and structures of two sulfa drugs of your syllabus. 2

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(b) What is pro-drug? Write down the name and structure of a pro-drug of your syllabus. 1.5

(c) Draw the structures of paracetamol and aspirin and answer the following questions: 2 2

(i) Write down the *IUPAC* name of them 1

(ii) How can you chemically distinguish between them in your laboratory? Give necessary chemical equations. 2 .

(iii) Give one medicinal use of aspirin and paracetamol. 2 .

(iv) What is baby aspirin? 1

Give a synthesis of aspirin starting from phenol 2.5 2.5

7. (a) Show a synthetic pathway for the preparation of 2-pentanone from ethyl acetoacetate (acetoacetic ester). 3

(b) Give a mechanism and one synthetic application for each of the following: 4×2

(i) Reimer-Tiemann reaction; (ii) Perkin reaction

(c) Write down the factors that affect Diels-Alder reaction with examples. 3

8. (a) Draw the structure of pyrrole, furan and thiophen with the numbering of all ring atoms and give answer to the following questions 3

(i) Which one of these three heterocyclic compounds has more canonical forms? Draw all possible canonical forms of those heterocyclic compounds. 3

(ii) What is the hybridization of the nitrogen atom of pyrrole? 1

(iii) In which orbital the lone pair electrons of the nitrogen atom of pyrrole are occupied? 1

(iv) Is pyrrole a basic compound? Explain. 2

(v) Is pyrrole an aromatic compound? If so, then why? 2

(v) Draw the orbital picture of pyrrole. 2

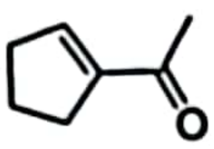
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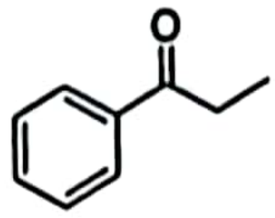
Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-II Examination 2022
Course Title: Organic Chemistry II (*Fundamentals of Organic Chemistry*)
Course: Chem. 230, Full Marks: 70, Time: 4 hrs.
[Answer any FIVE questions]

1. (a) Aldehyde or ketone reacts with 2° amine to give an enamine, but benzaldehyde cannot form an enamine with a 2° amine - Explain why? 2.0

(b) Which of the following carbonyl compounds does not undergo aldol condensation reaction and why? 2.0

(i) 

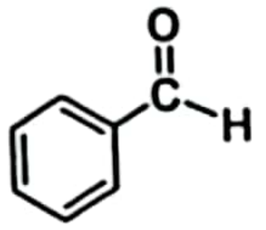
(ii) $(CH_3)_3C-\overset{O}{\overset{\|}{C}}-H$

(iii) 

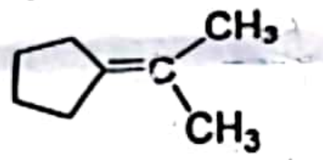
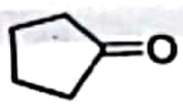
(c) Starting with benzaldehyde, outline the mechanistic steps for the synthesis of benzalacetone. 3.0

(d) Predict the products from each of the following reactions with explanation: 4.0

(i)  $\xrightarrow[AlCl_3]{CH_3CH_2COCl}$ A $\xrightarrow[HCl,\ H_2O]{Zn(Hg)}$ B

(ii)  $\xrightarrow[10\%\ NaOH,\ 5\,^\circ C]{H_3C\overset{O}{\overset{\|}{C}}H}$ C $\xrightarrow[(-H_2O)]{heat}$ D

(e) Show with mechanism how you would use a Wittig reaction for the following synthesis: 3.0

 from 

2. (a) *p*-Methoxybenzoic acid has pK_a of 4.46, and *p*-nitrobenzoic acid has pK_a of 3.41. Which acid is stronger? Give explanation. 2.0

(b) Show the product formed when 4-methylbenzoic acid is treated with each of the following reagents: 4.0

(i) $SOCl_2$; (ii) CH_3CH_2OH, HCl; (iii) NaOH, then CH_3I; (iv) $LiAlH_4$

(c) When methylacetate is heated in pure ethanol containing a small amount of HCl catalyst, ethyl acetate results:

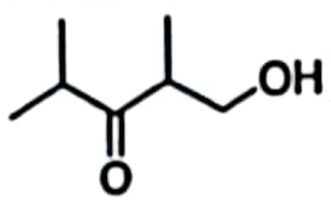
$$H_3C-\overset{O}{\overset{\|}{C}}-OCH_3 + CH_3CH_2OH \xrightarrow{H^+} H_3C-\overset{O}{\overset{\|}{C}}-OCH_2CH_3 + CH_3OH$$

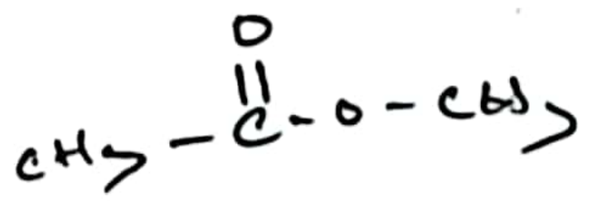
(i) What is the name of the reaction? 0.5

(ii) Show the mechanism of formation of ethyl acetate. 3.0

(iii) What happens when the hydrolysis of methyl acetate is carried out using H_2O^{18}? 1.5

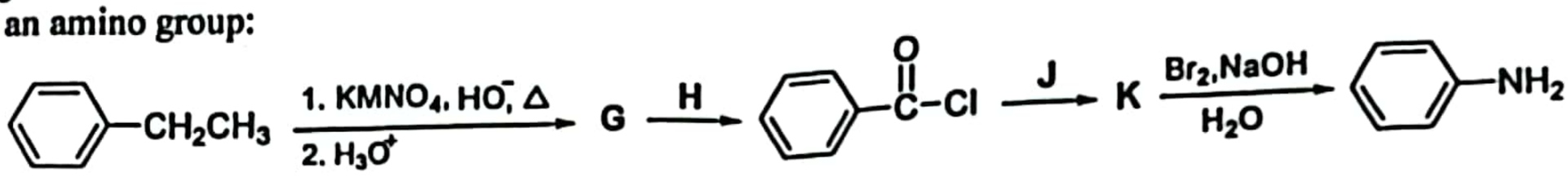
(d) Give structures for **E** and **F** in the following reaction with explanation: 3.0

 $\xrightarrow{H_2CrO_4}$ **E** $(C_7H_{12}O_3)$ $\xrightarrow{heat}$ **F** $(C_6H_{12}O)$ + CO_2



3. (a) Explain the fact that pyridine is less basic than piperidine. 2.0

(b) The reaction sequence below shows how an ethyl group on a benzene ring can be replaced by an amino group:

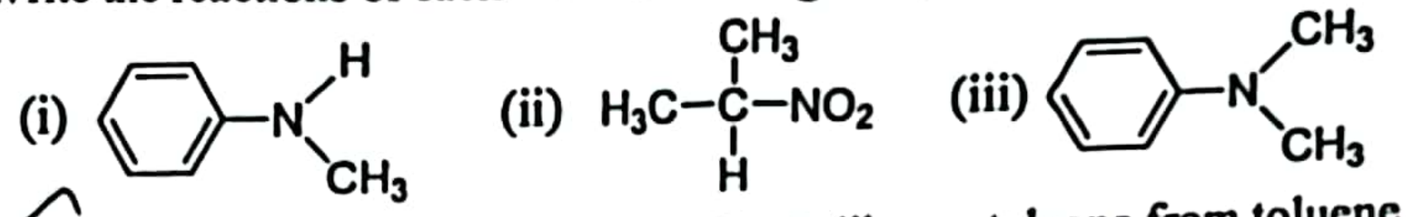


(i) Supply the missing reagents and intermediates. 3.0

(ii) Give a detailed mechanism for the formation of aniline from the compound K. 1.0

(iii) Show how you might prepare aniline from benzene. 3.0

(c) Write the reactions of each of the following compounds with nitrous acid:



3.0

(d) Outline all the steps in the synthesis of 3, 5-dibromotoluene from toluene. 1+2

4. (a) What are Barbiturates? Show the steps for the preparation of Barbital from diethyl malonate. 3×3

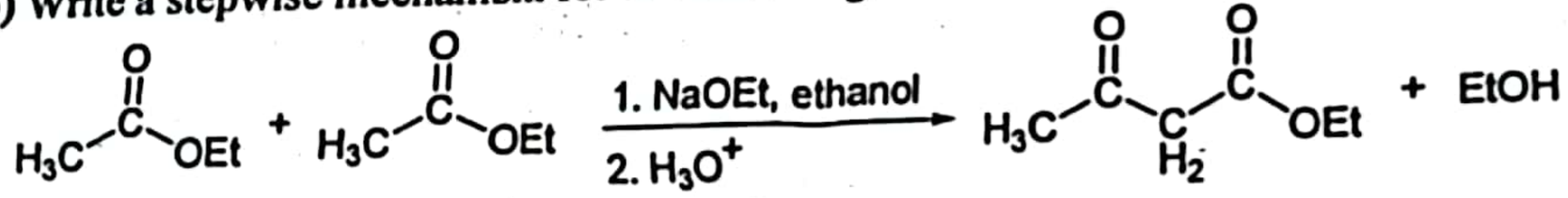
(b) Devise a synthesis of each of the following drugs:

(i) Aspirin from phenol; (ii) Saccharin from toluene; (iii) Sulfanilamide from benzene 2.0

(c) What kind of drug is plasmoquine? Draw its structure. 2.0

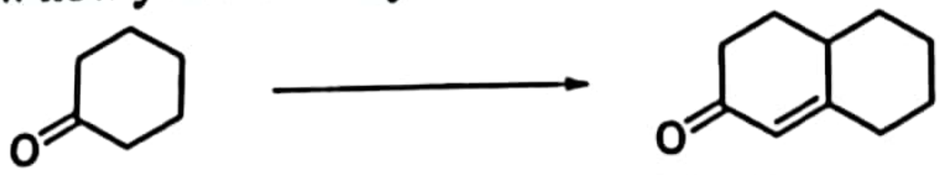
5. (a) Outline a reaction sequence for synthesis of 2-heptanone from ethyl acetoacetate. 3.0

(b) Write a stepwise mechanism for the following reaction:



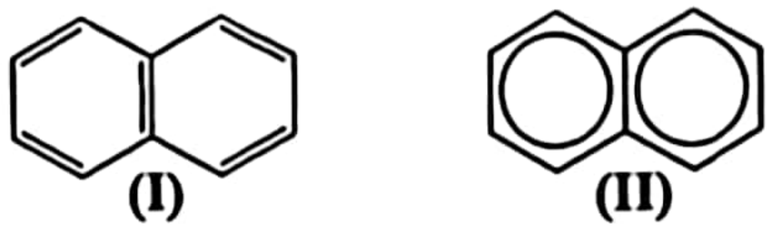
(c) Write the mechanism and two synthetic appiications for Perkin reaction. 3+2

(d) Show how you will carry out the following conversion: 2.0



(e) What are the factors favoring Diels-Alder Reaction. 2.0

6. (a) The following two structures can be drawn for the naphthalene. As a student of Chemistry which one the structures would you prefer to draw? Give explanation in support of your answer. 1.5



(b) Draw all the possible resonance structures of naphthalene and answer the following questions: 1.5

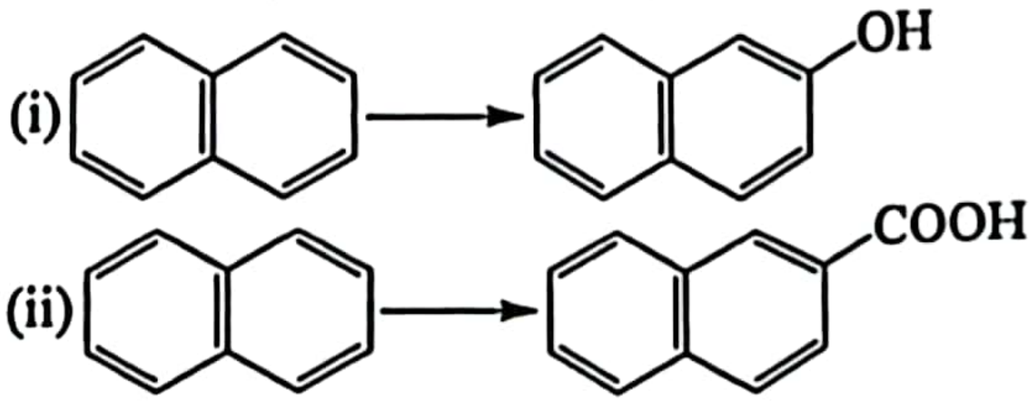
(i) Which one of the resonance structures that you have drawn is more stable? Give explanation in favor of your answer. 1.5

(ii) Which of the resonance structures are more reactive and why? 1.0

(iii) What is Fries rule? 1.0

(c) What is peri interaction? Give example. 1.0

(d) How would you carry out the following conversions? 2.5+4.0 = 6.5



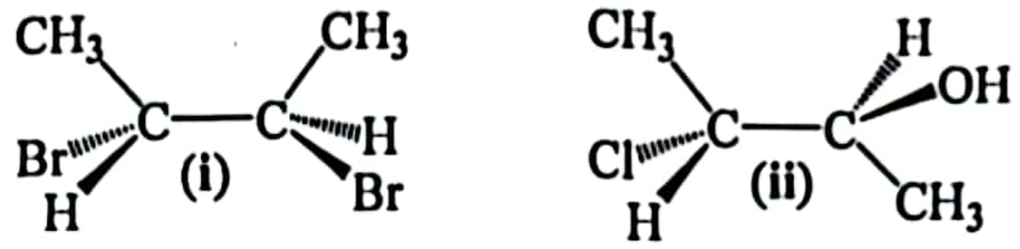
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7. (a) What are the basic differences between the constitutional isomer and stereoisomer? Give examples. 2.0

(b) Is there any basic difference between the chiral center and stereocenter? If so, then what is the difference? Explain with suitable examples. 3.0

(c) What is *meso* compound? Why is it so called? Explain with examples. 2.0

(d) Convert the following Wedge-dash projection into the Fischer Projection and assign the *R/S* configuration for each of the chiral carbon atoms on the basis of the Fischer Projection. 4.0



(e) "A Fischer projection formula can only be rotated at 180° on the plane of the paper but not at 90° or 270°"- Explain the statement with proper examples. 3.0

8. (a) Draw the structures of indole, benzofuran and benzothiophene with the numbering of all ring atoms and answer the following questions: 3.0

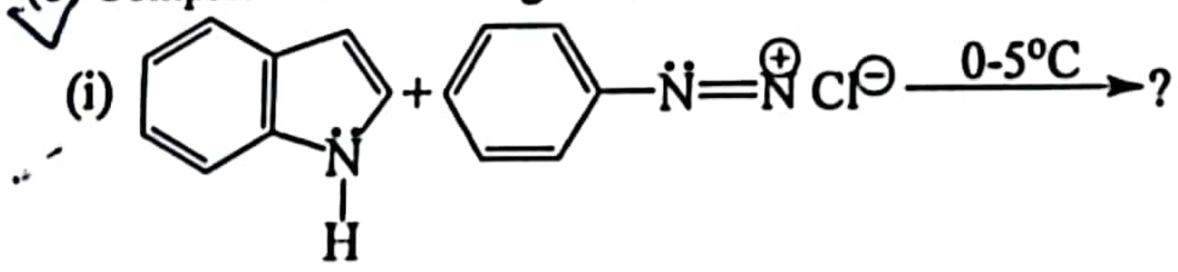
(i) Arrange them according to their decreasing order of reactivity and give reason in support of your arrangement. 2.0

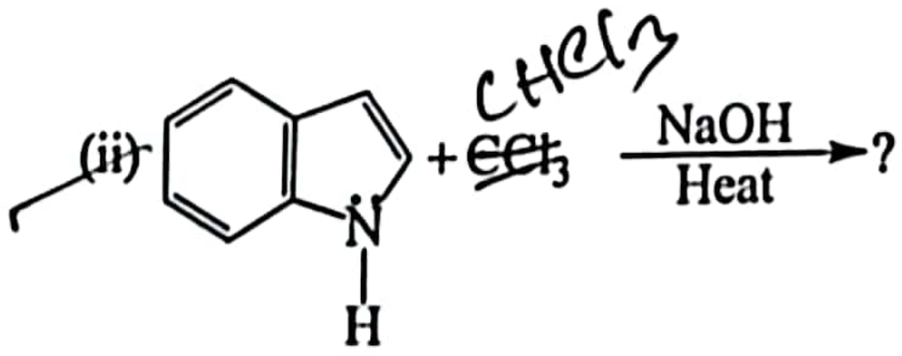
(ii) Why indole, benzofuran and benzothiophene are called the "*benzo*" derivative of benzene? 1.0

(iii) Why the electrophilic aromatic substitution reactions of indole, benzofuran and benzothiophene take place on the five membered heterocyclic rings? 2.0

(iv) "Electrophilic aromatic substitution reactions of benzothiophene takes place about equally at the positions C_2 and C_3 with the position C_3 being somewhat more reactive"-How can you *explain the statement?* 3.0

(b) Complete the following reactions: 3.0

(i) 

(ii) 

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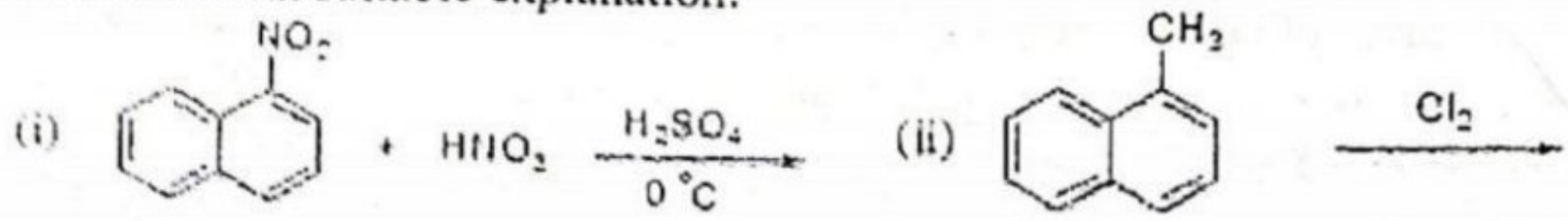
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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-II Examination 2021
Course Title: Organic Chemistry II (*Fundamental of Organic Chemistry*)
Course: Chem. 230, Full Marks: 70, Time: 4 hrs.
[Answer **any FIVE**questions]

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1. (a) Explain the conditions under which naphthalene undergoes sulfonation and Friedel-Crafts acylation at either α- or β-positions. 4.0

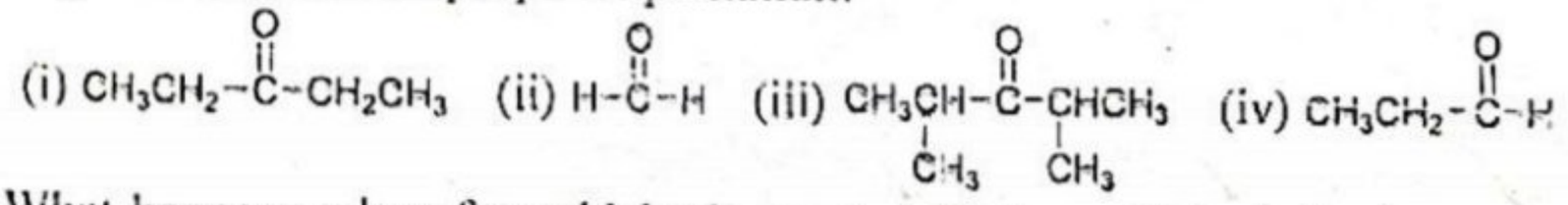
(b) Predict the orientation of electrophilic substitution in each of the following reactions with suitable explanation: 2+2



(c) Give the steps for the synthesis of each of the following compounds: 2+2
(i) 2-Naphthylamine from Naphthalene
(ii) Anthracene from benzene and phthalic anhydride

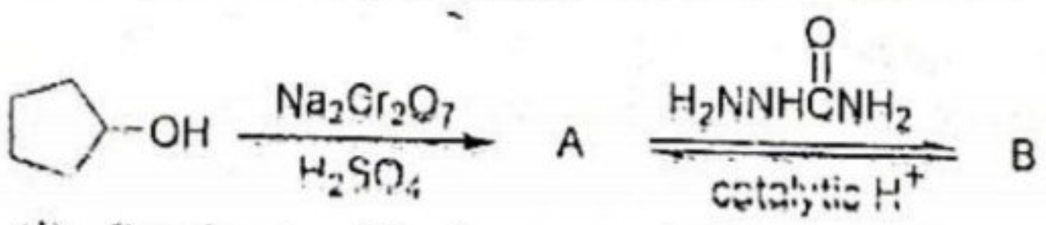
(d) Explain why phenanthrene undergoes oxidation and reduction more easily than naphthalene. 2.0

2. (a) List the following carbonyl compounds in order of decreasing reactivity toward nucleophilic attack with proper explanation: 2.0



(b) What happens when formaldehyde reacts with benzaldehyde in the presence of conc. NaOH? 2.0

(c) Draw the structures of the compounds (A and B) of the following reactions and answer the related questions: 1.0

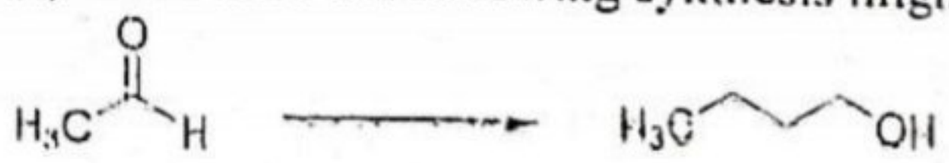


(i) Semicarbazide has two NH_2 groups. Explain why only one of them is involved in the formation of semicarbazone. 2.0

(ii) What product is formed when the compound A reacts with CH_3MgBr, followed by the addition of dilute acid? 1.0

(d) The compound (C) with molecular formula C_4H_8O does not give Tollens' test but gives positive iodoform test. Also the compound (C) on reduction with Zn(Hg)/conc. HCl give *n*-butane as product. Write the structure of (C) giving reasons for your answer. 3.0

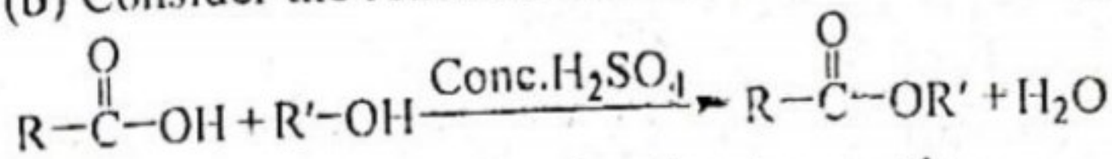
(e) Show how the following synthesis might be carried out: 3.0



3. (a) Why carboxylic acid is more acidic than phenol although the conjugate base of phenol gives more resonance structures than tha of the conjugate base of carboxylic acid? 2.0

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(b) Consider the reaction and answer the following questions:



(i) Propose a mechanism for the reaction. 3.0

(ii) Why is it necessary to use concentrated mineral acid, H_2SO_4 or dry HCl in the above esterification reaction? 1.0

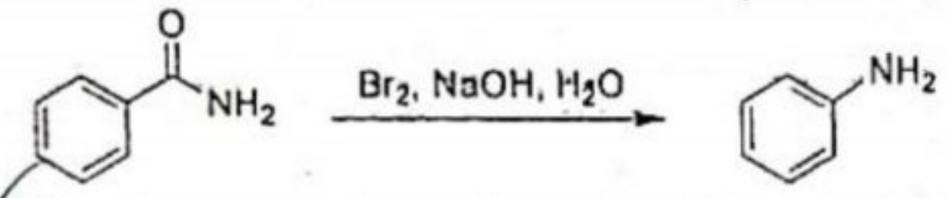
(iii) If we use secondary and tertiary alcohols instead of primary alcohol then what will be the rate of the above esterification reaction? Give explanation in favor of your answer. 3.0

(iv) Is the oxygen atom of the water molecule that liberated in the above reaction come from the carboxylic acid or alcohol? Give a chemical evidence in favor of your answer. 3.0

(c) How can you chemically distinguish between the formic acid and acetic acid? Give necessary chemical equation. 2.0

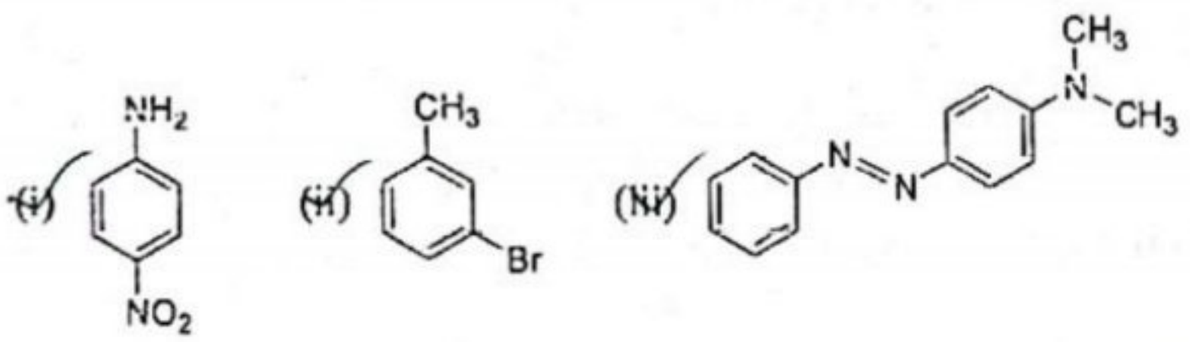
4. (a) Would you expect aniline to be more or less basic than cyclohexylamine? Why? 2.0

(b) Provide a detailed mechanism for the following reaction: 2.5



(c) What happens when nitroethane is treated with nitrous acid? 1.5

(d) Devise a synthesis of each of the following compounds startingfrombenzene: 3+ 3+2

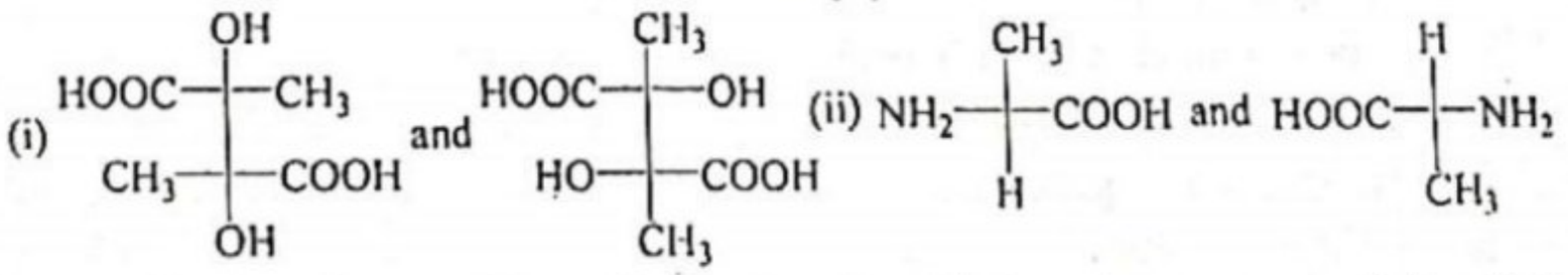


5. (a) What are the basic differences between the enantiomers and diastereomers? Explain with appropriate examples. 2.0

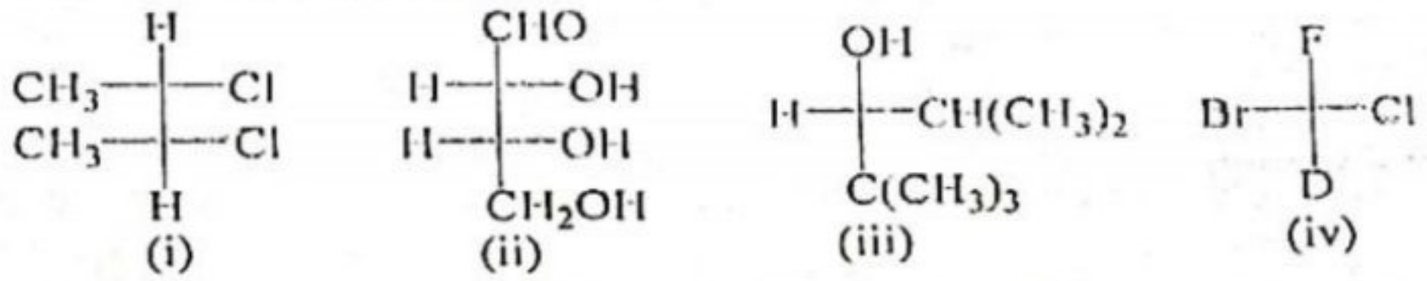
(b) What are the necessary conditions for an organic compound to be a diastereomer? Explain with suitable examples. 2.0

(c) What type of stereoisomers always obtain from a compound with one chiral carbon atom? Explain with example. 1.0

(d) Indicate whether the structures in each of the following pairs are enantiomers, diastercomers or identical molecules? Justify your answer. 3.0



(e) State whether the configuration of each chiral carbon atom of the following configurational stereoisomers is *R* or *S*. 4.0



(f) Define the term stereoisomers. How would you classify stereoisomers? 2.0

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6. (a) Paracetamol is a well-known antipyretic and analgesic drug. What do you understand by the terms antipyretics and analgesics? Show the steps for the preparation of paracetamol from phenol. 4.0

(b) Provide a method for the synthesis each of the following drugs: 2.5×4

(i) Prontosil; (ii) Dulcin;

(iii) Phenobarbital; (iv) Plasmoquine

7. (a) Show how you would use the acetoacetic ester synthesis to prepare 3-propyl-2-hexanone. 3.0

(b) Give a method for the preparation of benzalacetophenone. Propose a mechanism for the reaction of benzalacetophenone with hydrogen cyanide. 3.0

(c) Write the mechanism and one synthetic application for each of the followings: 3+3

(i) Diels-Alder reaction

(ii) Reimer–Tiemann reaction

(d) What two reactions are involved in Robinson annulation? Show the necessary reactions. 2.0

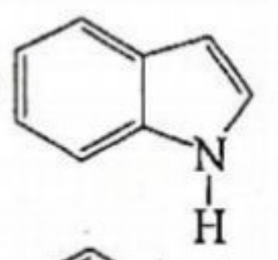
8. (a) "Quinoline and isoquinoline undergo electrophilic aromatic substitution reaction on the benzene ring but the nucleophilic substitution reaction take place on the pyridine ring"- How would you explain the statement? 4.0

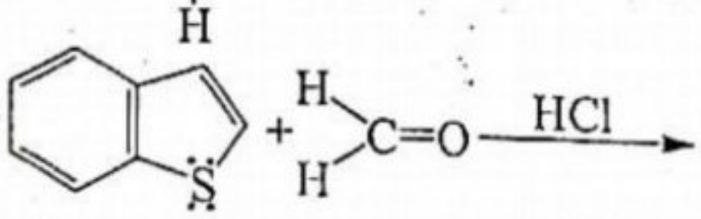
(b) Explain the basicity of imidazole. 2.0

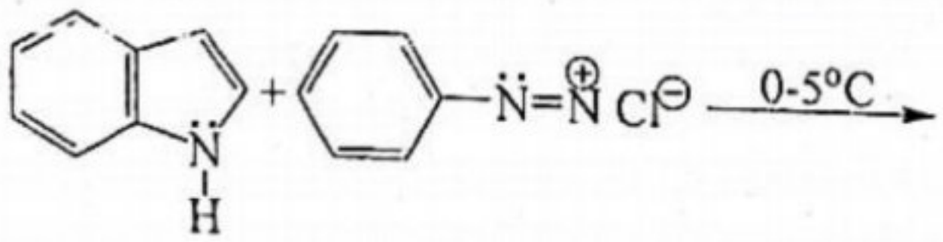
(c) Draw the orbital picture of imidazole showing different types of bonds. 2.0

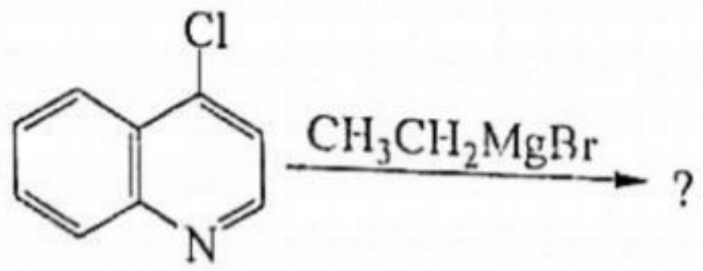
(d) "Imidazole is an aromatic compound"- How would you explain it? 2.0

(e) Write down the product(s) of the following reactions: 4.0

(i)  + Br_2 ⟶ ?

(ii) 

(iii) 

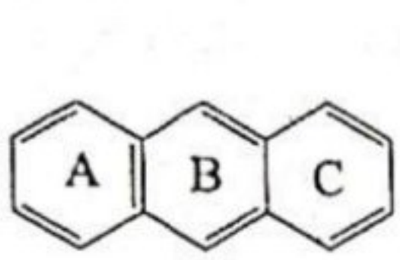
(iv)  CH_3CH_2MgBr ⟶ ?

=0=0=0=

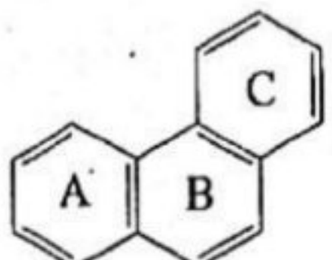
Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part - II Examination-2019
Course Title: Organic Chemistry II (Fundamental of Organic Chemistry II)
Course No: Chem-230 Musa 50
Full Marks: 70, Time: 4.00 hours

(Answer Any Five Questions)

1 (a) Number the all ring carbon atoms of the following two compounds (I and II) and answer the following questions: 1+2+2+1 +1+3



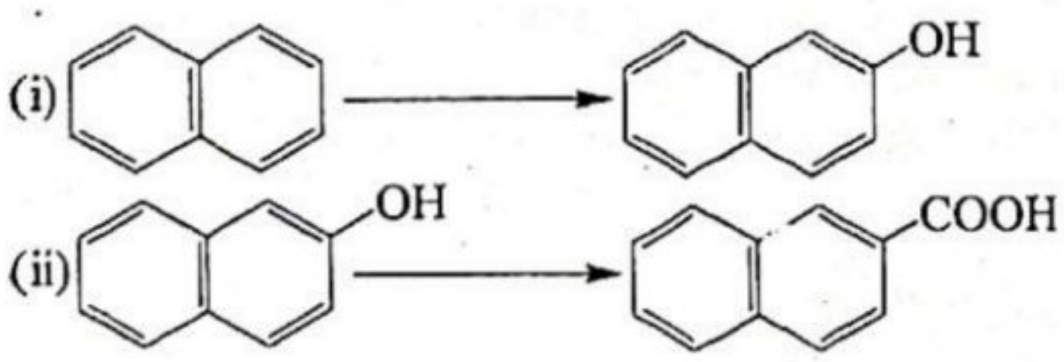
(I)



(II)

(i) Why the resonance energy of the compound (II, 92kcal/mol) greater than the compound (I, 84 kcal/mol) although both of them have three fused benzene rings system?
(ii) Why the electrophilic aromatic substitution reaction of the above two compounds (I & II) occurs in the middle ring B at the 9,10 positions instead of the rings A and C? Give explanation in support of your answer.
(iii) Which of the compounds (I &II) readily undergo reductive ozonolysis reaction? Give reason in favor of your answer.
(iv) Which of the compounds (I &II) will form adduct with maleic anhydride? Give explanation in favor of your answer.
(v) Write down the Haworth synthesis of the compound (II).

(b) How will you carry out the following conversions: 2+2



2 (a) Give the preparation of ketones from acid chlorides with organocopper compounds Why Grignard reagents are not used in this reaction instead of organocopper compounds? 3

(b) How will you carry out the following conversions? 4
(i) Acetophenone into α-phenyl ethyl alcohol
(ii) Benzaldehyde into benzylamine
(iii) Acetaldehyde into acetaldoxime
(iv) Benzoyl chloride into benzophenone.

(c) Explain the following: 2+2
(i) Aromatice aldehydes are less reactive than aliphatic aldehydes towards nucleophilic addition reaction.
(ii) The alpha hydrogen of a carbonyl compound is acidic.

(d) How are hemiacetals and acetals formed? Give mechanism. 3

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3. (a) What are the basic differences between the diastereomers and meso compounds? Give examples 1.5

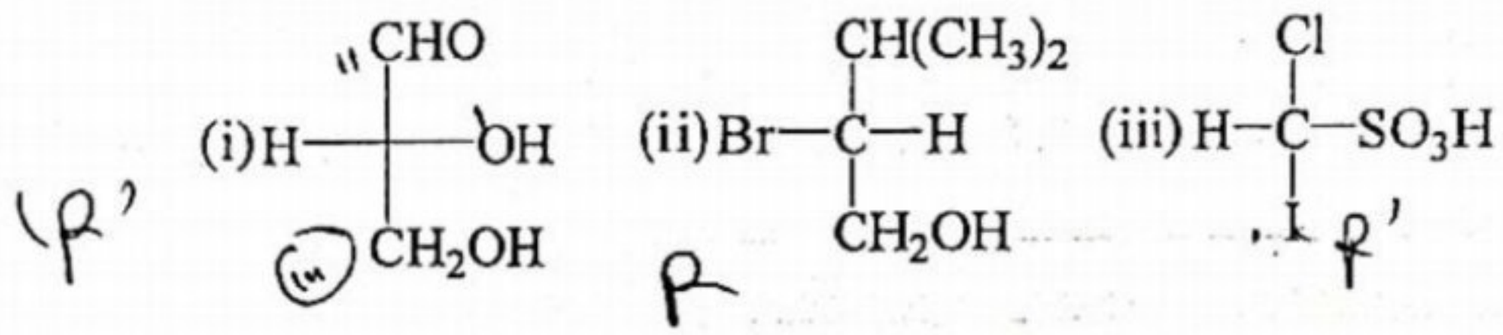
(b) Define the terms **"chiral center"** and **"pro-chiral center"** with suitable examples. 1.5

(c) Define specific rotation. 28.0 mg of mandelic acid was dissolved in 1.00 cm3 of ethanol and the solution placed in a 10.0 cm long polarimeter cell. An optical rotation α of - 4.35 was measured at 20°C with light of wave length 589 nm. Calculate the specific rotation of the mandelic acid. 1+2

(d) What is racemic mixture? How would you represent a racemic mixture? 1.5

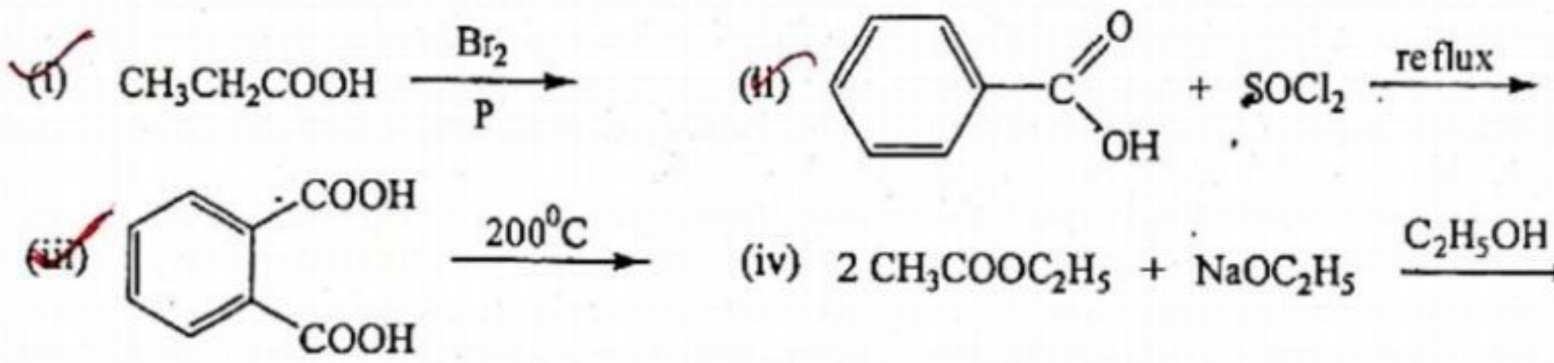
(e) What is meant by the term "resolution of a racemic mixture"? Discuss the resolution of racemic α-phenylethylamine, $C_6H_5CH(NH_2)CH_3$ into its individual enantiomers. 1+2.5

(f) Assign the R/S configuration of the following compounds: 3



4. (a) Describe how the inductive and the resonance effects influence the acidity of carboxylic acids. 4

(b) Complete the following reactions and indicate the name of the products: 1×4 = 4



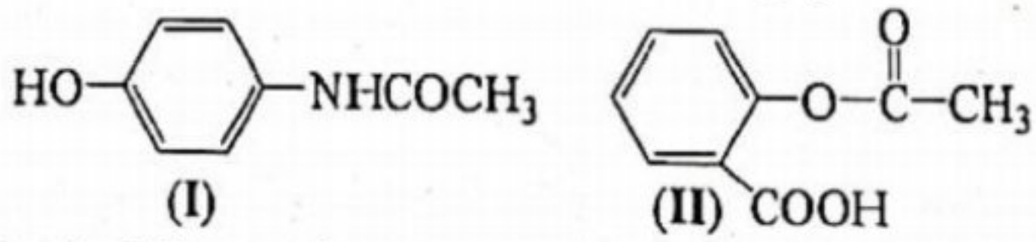
(c) "Hydrolysis of ester shows acyl-oxygen cleavage instead of alkyl-oxygen cleavage. Give two evidences in favor of this statement. 3

(d) How can you prepare β-hydroxycarboxylic acids by Reformatsky reaction? 2

(e) What is vinegar? 1

5. (a) What are the sulfa drugs? Write down the name, structure and one medicinal use of two sulfa drugs. 1+2

(b) Consider the structures (I&II) and answer the following questions: 1+2+1+3



(i) What is the basic difference between the two drugs (I & II).
(ii) How can you chemically distinguish between these two drugs? Give necessary chemical equation.
(iii) Give two medicinal uses of the drug (II).
(iv) Starting from phenol how can you synthesize the drug (II).

(c) What is Michael addition reaction? Give example. 1

(d) What type of nucleophiles are used in the Michael addition reaction? 0.5

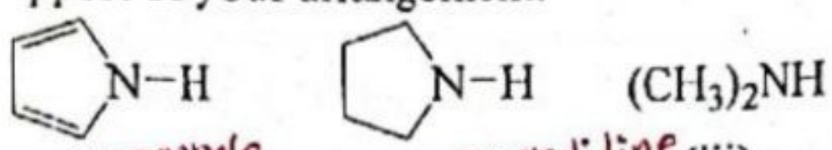
(e) Write down the mechanism of the Michael addition reaction. 2.5

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6 (a) What are amines? Give example 1

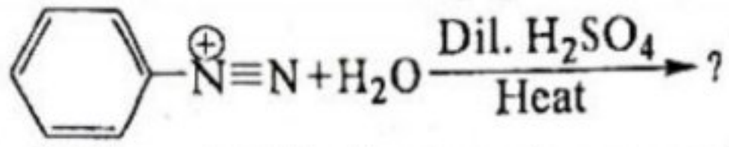
(b) Briefly discuss two different methods that only use for the preparation of primary amines. 6

(c) Arrange the following compounds according to their increasing order of basicity and give explanation in support of your arrangement. 3



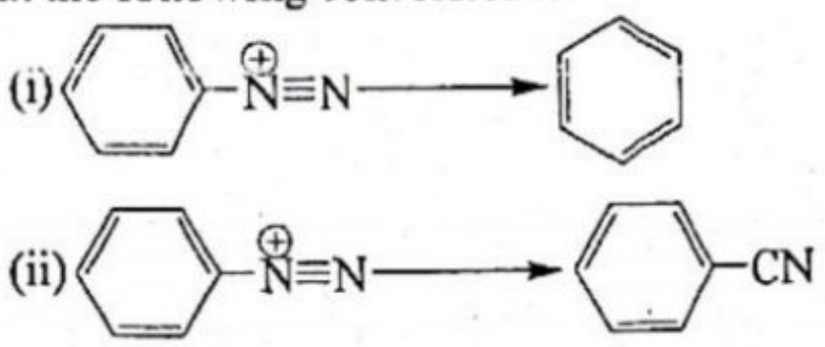
(i) (ii) (iii)

(d) Complete the reaction and answer the following question: 1+1

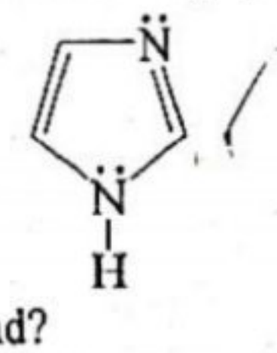


What is the function of dilute H_2SO_4 in the above reaction? Give explanation in support of your answer.

(e) How can you carry out the following conversions: 2

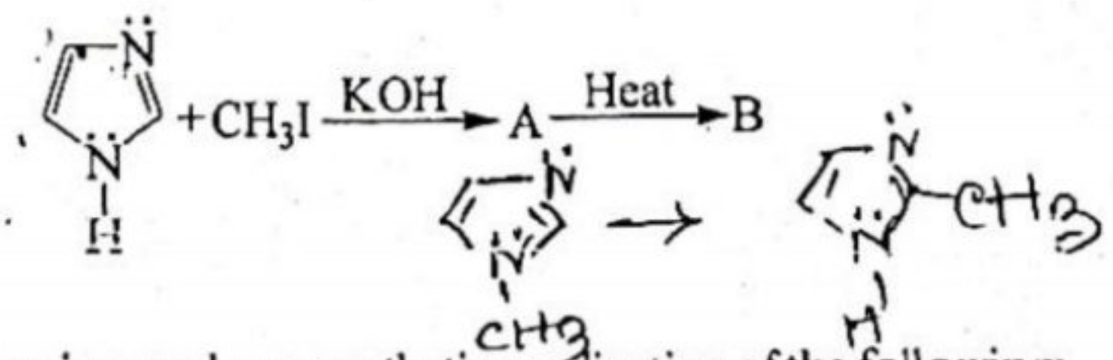


7 (a) Consider the structure and answer the following questions: 0.5+0.5+3+2+2+2+2



(i) What is the name of the compound?
(ii) Number all the ring atoms of the molecule.
(iii) Draw the orbital picture of the compound.
(iv) Is the compound aromatic? Give explanation in favor of your answer.
(v) Which of the nitrogen atoms of the compound is more basic? Give explanation.
(vi) "The compound is more reactive than pyridine but less reactive than pyrrole towards the electrophilic aromatic substitution reaction"- How would explain the statement.
(vii) Give a synthesis of the compound.

(b) Draw the structures of the unknown compound A&B in the following series of reactions. 2



8 (a) Give the reaction, mechanism and one synthetic application of the following: 3+3+3+3

(i) Cannizzaro reaction
(ii) Reimer-Tiemann reaction
(iii) Perkin reaction
(iv) wittig reaction

(b) Discuss nuicleophilic addition reaction for α, β-unsaturated carbonyl compounds. 2

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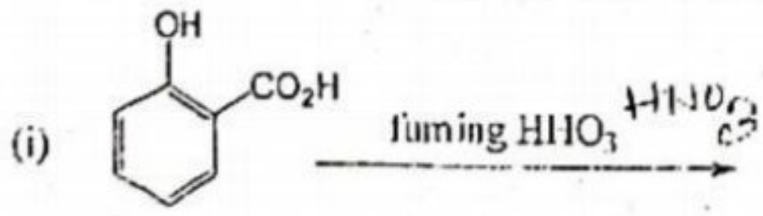
Department of Chemistry
Jahangirnagar University
B.Sc (honors) Part-II, Final Examinations-2018
Course Title: Organic Chemistry II (Fundamentals of Organic Chemistry II)
Course No.: Chem-230, Total Marks: 70, Time 4.0 hours Musa 50

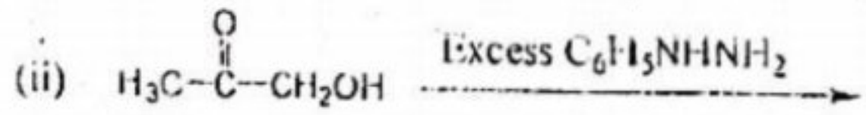
(Answer any five questions)

1. (a) Draw the structures of anthracene and phenanthrene with numbering all carbon atoms and answer the following questions: 2.0
   (i) Are anthracene and phenanthrene bridgew or fused ring polycyclic compounds? 0.5
   (ii) Are anthracene and Phenanthrene aromatic compounds? Give explanation in support of your answer. 2.0
   (iii) Are all the carbon-carbon bonds in anthracene and phenanthrene same. Give reason in support of your answer. 3.5
   (iv) How can you chemically distinguish between anthracene and phenanthrene? Give necessary equations. 2.0
   (b) Give a method for the preparation of anthracene. 4.0

2. (a) What are the necessary conditions for an organic compound to exhibit diastereomers and meso compounds? Explain with appropriate examples. 4.0
   (b) Draw the theoretically possible all configurational stereoisomers of the compound $CH_3CH(OH)CH(OH)CH_2CH_3$ and answer the following questions(give explanation where necessary): 2.0
   (i) Assign (R)/(S) configuration of each configurational isomer. 4.0
   (ii) How many enantiomeric and diastereomeric pairs are there? 3.0
   (iii) Is there any meso compound? 1.0

3. (a) How would you synthesize the following drugs? 5x2 = 10
   (i) Sulphadiazine (ii) Mepacrine (iii) Paracetamol (iv) Barbital (v) Dulcin
   (b) How can you prepare β-hydroxyl carboxylic acid by Reformatsky reaction? 2.0
   (c) Complete the following reactions indicating the name of the products. 2.0





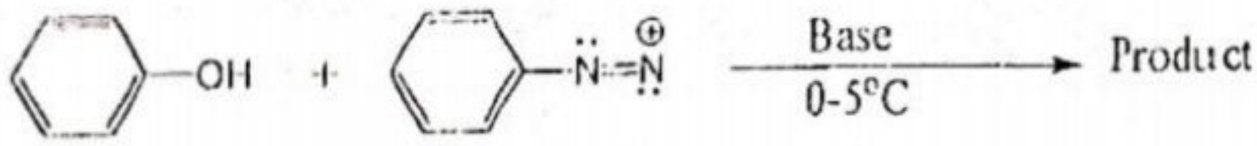
4. (a) What are amines? Give an example of a primary aromatic and a primary aliphatic amines with structure and answer the following questions: 2.0
   (b) Which of the amines is more basic? Give explanation in support of your answer. 2.0

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(c) How will you distinguish them chemically? Give the necessary chemical equations. 2.0

(d) Give a synthesis of each of them. 4.0

(b) How would pH of the reaction medium affect the rate of following coupling reaction?



(i) If the pH of the reaction medium is <7 0.5

(ii) If the pH of the reaction medium is 7 0.5

(iii) If the pH of the reaction medium is >7 but <10 0.5

(iv) If the pH of the reaction medium is > 10 2.0

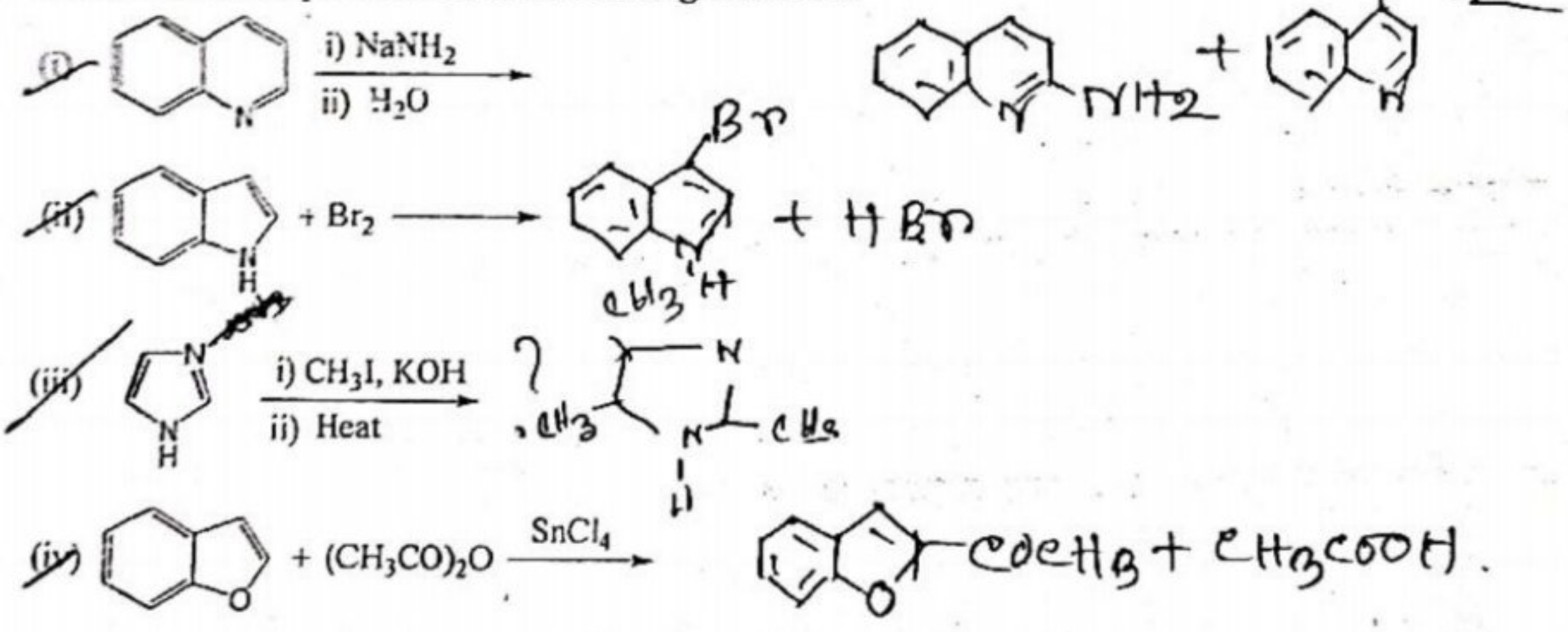
(v) Graphically represent the role of pH on the above reaction.

5.(a) "Quinoline and isoquinoline undergoes electrophilic aromatic substitution reaction on the benzene ring but nucleophilic substitution reaction takes place on the pyridine ring"- How would explain the statement? 4.0

(b) Explain the basicity of imidazole. 2.0

(c) "Imidazole is an aromatic compound"- How would you explain it? 1.0

(d) Write down the product of the following reactions: 4.0

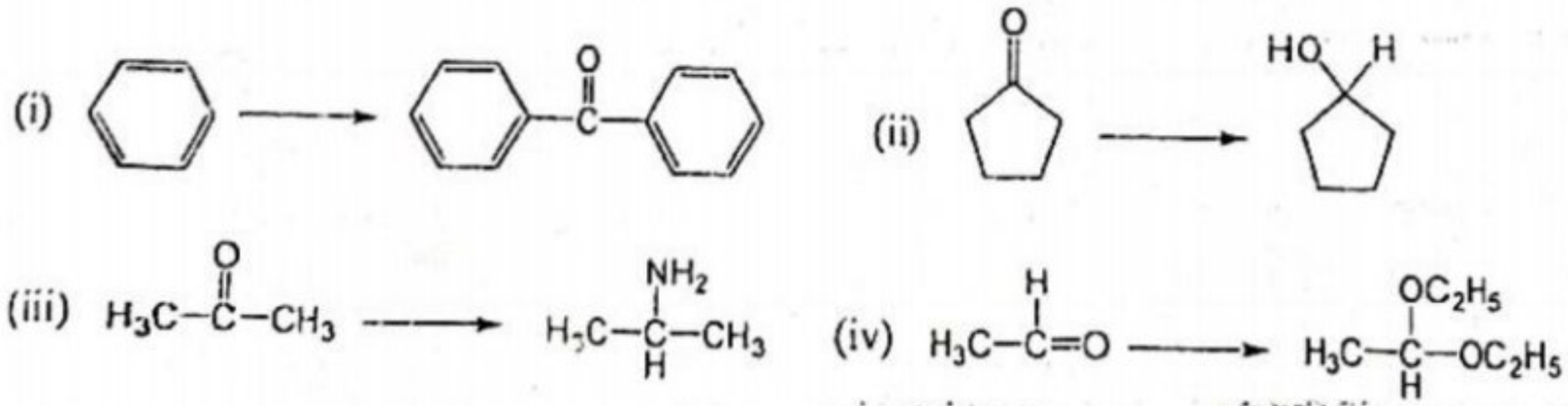


(e) Give a method for the preparation of imidazole. 3.0

6. (a) Give two methods of preparation of aldehyde. 3.0

(b) Write down the general mechanism of acid catalyzed nucleophilic addition to carbonyl compounds. 2.0

(c) How will you carry out the following conversions: 4.0



(d) How can you prepare different types of alcohols from carbonyl compounds using Grignard reagent? 3.0

(e) Explain the fact that the α-hydrogen of carbonyl compound is acidic? 2.0

7. (a) Arrange the following compounds in decreasing order of their acid strength and give reason in support of your answer. 2.0

CH_3CH_2COOH, $H_2C{=}CH\text{-}COOH$, $CH_3\text{-}COOH$ and $H\text{-}COOH$.

(b) Give a method for the preparation of α,β-unsaturated acid. 3.0

(c) "α,β-unsaturated acids react with HBr in the absence of any peroxide according to the anti-Markonikov's rule"- How would you explain the statement? 2.0

(d) Write down the product and mechanism of the following reactions: 3+2 = 5.0

(i) $R{-}\overset{\overset{O}{\|}}{C}{-}OH + CH_2N_2 \longrightarrow R{-}\overset{\overset{O}{\|}}{C}{-}OCH_3 + N{\equiv}N$

(ii) $R{-}\overset{\overset{O}{\|}}{C}{-}OH + SOCl_2 \longrightarrow R{-}\overset{\overset{O}{\|}}{C}{-}Cl + SO_2 + HCl$

(e) Complete the following reactions: 2.0

$HOCH_2(CH_2)_4CH_2COOH$ → (H^+, Heat; Dilute Solution) / (H^+, Heat; Concentrated Solution)

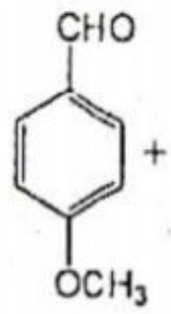
8. (a) Give the reaction, mechanism and one synthetic application of the following: 3x4 =12

(i) Aldol condensation (ii) Michael addition

(iii) Benzoin condensation (iv) Hoffman degradation

(b) Complete the following reactions: 2.0

(i) $H_3C{-}\overset{\overset{O}{\|}}{C}{-}CH_3 \xrightarrow[H_2SO_4]{NaCN} ? \xrightarrow[\Delta]{H_2SO_4} ?$

(ii)  + HCHO $\xrightarrow{\text{Conc. NaOH}}$? + ?

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-II Examination 2017
Course Title: Organic Chemistry II
Course: Chem. 230
Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) What is iodoform reaction? What type of compounds can give this reaction? 3.0
(b) "$>C=O$ gives nucleophilic addition reaction, whereas $>C=C<$ undergoes electrophilic addition" – Explain. 3.0
(c) How would you synthesize the followings: 4.0
(i) Acetophenone from benzene
(ii) Acetaldoxime from acetaldehyde
(iii) Cyclopentane from cyclopentanone
(iv) Mandelic acid from benzaldehyde
(d) An alkene, C_6H_{12}, after ozonolysis yielded two products. One of these gave a positive iodoform reaction but a negative Tollens test. The other product gave a positive Tollens test, but a negative iodoform reaction. Deduce the structure and give all necessary equations. 4.0

2. (a) Define enantiomer and diasteroismer. Give examples. 3.0
(b) Draw all the stereoisomers of 2,3-butane-diol, $CH_3CH(OH)CH(OH)CH_3$. 4.0
(c) What is resolution? Give a method of resolving a recemic mixture. 3.0
(d) Define specific rotation & explain why is plane-polarized light used in the analysis of enantiomers? 4.0

3. (a) Give the Haworth synthesis of naphthalene. 3.0
(b) What happens when naphthalene is reacted with- 4.0
(i) Conc. H_2SO_4 at 160 °C (ii) Na, $C_5H_{11}OH$, reflux
(iii) O_2, V_2O_5 at 460 °C (iv) CH_3COCl, $AlCl_3$ in nitrobenzene
(c) Explain the fact that anthracene and phenanthrene are less resistant towards oxidation and reduction than naphthalene. 4.0
(d) How do you account for the aromaticity of anthracene? What is the characteristic reaction of anthracene? 2+1 = 3

4. (a) How would you synthesize the following drugs: 5×2 = 10
(i) Aspirin (ii) Uramil (iii) Saccharine (iv) Sulphanilamide (v) Plasmoquine
(b) Give the mechanism of electrophilic addition of α,β-unsaturated carbonyl compounds. 4.0

5. (a) Give the mechanism and synthetic application of the followings: 3×4
i) Perkin reaction, ii) Knoevenagel and Stobbe Reaction,
iii) Diels-Alder reaction, iv) Wittig reaction.
(b) How will you prove that Cannizzaro reaction involves a hydride transfer? 2.0

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(a) Give a method of synthesis for each of the following: 2×
i) Indole, ii) Isoquinoline, iii) Pyrimidine 1×3
(b) Predict the products of reaction of indole with the following reagents:
i) SO_3, pyridine, Δ ii) CH_3I, DMF, Δ iii) HCHO, $(CH_3)_2NH$ 2×3 = 5
(c) Discuss the Skraup synthesis of quinoline and give the mechanism of its formation.

4.0
Y(a) Give two methods of synthesis for each of the following:
i) Primary amine, ii) Secondary amine
(b) Discuss the basicity of $ArNH_2$, R_2NH and RNH_2 and show their order of basicity. 3.0
(c) Why is direct nitration of aniline difficult? 2.0
(d) How can you introduce –F and –COOH in benzene ring via diazotization? 2.0
(e) Give the mechanism of formation of diazonium salt from primary aromatic amine. 3.0

3. (a) Why does an acyl chloride undergo nucleophilic attack more rapidly than does an alkyl chloride? 3.0
(b) Give the products of the reactions of $LiAlH_4$ with 5.0
i) $PhCH_2COOCH_3$ ii) $(CH_3CO)_2O$ iii) PhCOCl

2016

iv) v) $(CH_3)_2CHCH_2CN$

(c) How can you prepare α-hydroxycarboxylic acids and α,β-unsaturated carboxylic acid? 3.0
(d) Acidity of carboxylic acids depends on nature of the substituents - Explain. 3.0

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-II Examination 2016
Course Title: **Organic Chemistry II**
Course # Chem. 230, Full Marks: 70, Time: 4 hours

[Answer any FIVE questions]

1. (a) What are polynuclear hydrocarbons? Discuss the structure of naphthalene. 1+3
(b) All carbon-carbon bonds in naphthalene are not of the same length – Explain. 2
(c) Give the Haworth synthesis of phenanthrene. 3
(d) Carry out the following conversions: 1×4
(i) α-Naphthylamine from naphthalene.
(ii) 9.10-dihydrophenanthrene from phenanthrene.
(iii) 2-Naphthylamine from 2-napthol.
(iv) 9-Bromoanthracene from anthracene.
(e) How do you account for the aromaticity of anthracene?

2. (a) What are the necessary and sufficient conditions for optical activity of an organic molecule? 2
(b) How many stereoisomers of 2,3-butanediol are possible? Draw them. 2+2
(c) Differentiate between enantiomers and diastereomers. 2
(d) Reaction of 2-butanone with HCN yields a chiral product. What stereochemistry does the product have? Is it optically active? 2
(e) What is racemic mixture? Give at least two methods to prepare a racemic mixture. 2

3. (a) Discuss how is chemistry of aldehydes and ketones governed by carbonyl group? 4.5
(b) An optimum pH is necessary for running the reaction between ammonia derivatives and carbonyl compounds steadily – explain. 4.5
(c) How would you carry out the following conversions? 5
(i) Cyclopentane from cyclopentanone.
(ii) Acetaldehyde diethyl acetal from acetaldehyde.
(iii) Dibenzal acetone from benzaldehyde.
(iv) Cyclohexylamine from cyclohexanone.
(v) Acetophenone from benzene.

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4. (a) Give the mechanism and one synthetic application of the following: 3×4
(i) Dields-Alder reaction, (ii) Benzoin condensation, (iii) Reimer-Tiemann reaction, (iv) Wittig reaction.

(b) Acetoacetic ester is an equilibrium mixture of the keto and enol form. Give evidence in support of the statement. 2

5. (a) How would you synthesize the following drugs? 2×5
(i) Sulphadiazine, (ii) Prontosil, (iii) Paracetamol, (iv) Phenobarbital (v) Dulcin.

(b) Discuss the use of active methylene compounds in organic synthesis. 2

(c) Why do carboxylic acid derivatives undergo nucleophilic substitution reactions readily?

6. (a) Give a method of synthesis for each of the following: 2×3
(i) Indole, (ii) Isoquinoline. (iii) Pyrimidine.

(b) Predict the products of reaction of indole with the following reagents: 1×5
(i) SO_3, pyridine, Δ, (ii) CH_3I, DMF, Δ, (iii) HCHO, $(CH_3)_2NH$, (iv) Sn. HCl
(v) O_3, $HCONH_2$

(c) Discuss the structure and chemistry of imidazole. 3

7. (a) Give two methods of synthesis for each of the following: 4
(i) Primary amine (ii) Secondary amine

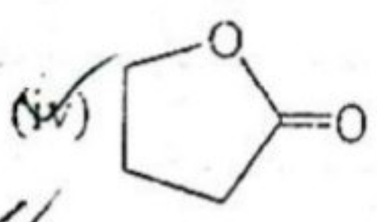
(b) Discuss the basicity of $ArNH_2$, R_2NH and RNH_2 and show their order of basicity. 3

(c) Why is direct nitration of aniline difficult? 2

(d) How can you introduce –F and –COOH in benzene ring via diazotization? 2

(e) Discuss the reaction of HNO_2 on different nitroalkanes. 3

8. (a) Why does an acylchloride undergo nucleophilic attack more rapidly than does an alkylchloride? 3

(b) Give the products of the reactions of $LiAlH_4$ with 5
(i) $PhCH_2COOCH_3$, (ii) $(CH_3CO)_2O$, (iii) PhCOCl
(iv)  (v) $(CH_3)_2CHCH_2CN$

(c) Give the mechanism of alkaline hydrolysis of esters. 3

(d) Acidity of carboxylic acids depends on nature of the substituents – Explain. 3

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part - II Examination-2015
Course Title: Organic Chemistry II (Fundamental of Organic Chemistry II)
Course No: Chem-230
Full Marks: 70, Time: 4.00 hours

(Answer Any Five Questions)

1. (a) Synthesize the following compounds: 2+2
(i) Phenanthrene (ii) Sulphanilamide
(b) α -position of naphthalene is more active than β --position towards electrophilic aromatic substitution-explain. 3
(c) What happens when naphthalene is treated with 4
(i) Na/C_2H_5OH reflux (ii) $Na/C_5H_{11}OH$ reflux (iii) H_2SO_4, 80°C (iv) H_2SO_4, 160°C
(d) Show the numbering of anthracene. What is the characteristic reaction of anthracene? Explain that the reaction occurs readily at 9-position. 3

2. (a) The characteristics reaction of carbonyl compounds are nucleophilic addition whereas acid derivatives shoe nucleophilic substitution –explain it in details. 3+3
(b) Acetal and ketal formation is pH dependent-explain. 3
(c) Give the reactions for the preparation of 1°, 2° and 3° alcohols from carbonyl compounds. 3
(d) Write down the reaction by which acetaldehyde and acetone can be distinguished. 2

3. (a) What is esterification reaction? Show the reaction for the formation of ethyl acetate by esterification reaction. 1+1
(b) How the yield of an esterification reaction can be increased? Write down the different possibilities. 2
(c) Show the mechanism for acid -catalyzed esterification. 2
(d) What is Hell-Volhard-Zelinsky reaction? How you can prepare α-hydroxycarboxylic acids and α, β -unsaturated carboxylic acids using this reaction? 1+3
(e) Arrange the acidity of the following acids in increasing order. Give the explanation in support of your answer. 4
(i) Acetic acid (ii) Trichloroacetic acid (iii) Benzoic acid (iv) *o* and *p*-Nitrobenzoic acid (v) Propanoic acid.

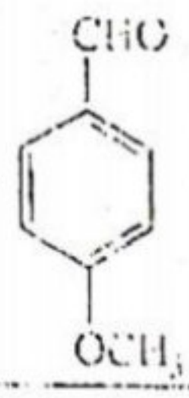
4. (a) Give an industrial method of the preparation of aniline. 2
(b) How can you differentiate 1°, 2° and 3° amines by a laboratory test? 3
(c) Show the reactions of HNO_2 on different types of amines. 3
(d) Outline a synthesis of *m*-bromotoluene from toluene. 3
(e) Write down a method to prepare tertiary nitro alkanes with example. 1.5
(f) Show the reduction of nitrobenzene in alkaline medium. 1.5

5. (a) Give the reaction, mechanism and one synthetic application of the following: 3×4
(i) Aldol condensation (ii) Perkin reaction (iii) Benzoin condensation (iv) Hofmann degradation.

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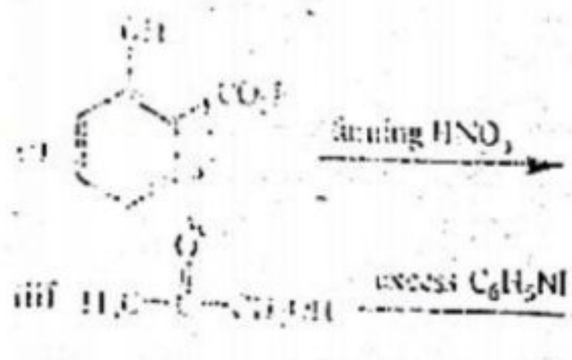
(b) Complete the following reaction:

(i) $H_3C-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[H_2O]{NaCN} ? \xrightarrow[\Delta]{H_2SO_4} ?$

(ii) 

$+ HCHO \xrightarrow{\text{conc. NaOH}} ? + ?$

6. (a)	What is plane-polarized light? How can you produce it from ordinary light?	2
(b)	"Enantiomers have different physiological properties" -- explain.	2
(c)	Draw the possible stereoisomers of 2,3-dichloropentane and find out their relationship as enantiomer and diastereoisomer.	3
(d)	How does a *meso* compound differ from racemic modification? Illustrate with suitable examples.	3
(e)	Discuss the resolution of a racemic mixture based on the formation of diastereoisomers	4
7. (a)	How would you synthesize the following drugs: (i) Sulphadiazine (ii) Plasmoquin (iii) Aspirin (iv) Diuluric acid (v) Saccharin.	2×5
(b)	How can you prepare β-hydroxy carboxylic acids by Reformatsky reaction?	2
(c)	Complete the following reactions indicating the name of the products.	2



(a)	Give a method of synthesis for each of the following: (i) Indole (ii) Quinoline (iii) Imidazole.	1.5×3
(b)	Explain the following facts: (i) Positions 4 and 5 are equivalent in the imidazole ring. (ii) Indole undergoes electrophilic aromatic substitution at C-3 predominantly than C-2.	1.5+2
(c)	Predict the products of reactions of quinoline with the following reagents: (i) fuming H_2SO_4 (ii) $NaNH_2$ in H_2N (iii) alkaline $KMnO_4$ (iv) Sn, HCl.	4
(d)	Complete the following reactions:	1×2

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-II Final Examination, 2014
Course Title: Organic Chemistry II
Course: Chem. 230
Full Marks: 70, Time: 4 hours

[Answer any FIVE questions.]

1. (a) What is a polynuclear aromatic hydrocarbon? 1
(b) The C_1-C_2 bond length in naphthalene is 1.365 A° while that for C_2-C_3 is 1.404 A°. Account for this fact. 3
(c) Give the Haworth synthesis of phenanthrene. 1×4
(d) What happens when naphthalene is reacted with:
(i) Conc. H_2SO_4 at 160°C; (ii) Na, C_2H_5OH, reflux; (iii) O_2, V_2O_5 at 460°C and
(iv) CH_3COCl, $AlCl_3$ in nitrobenzene. 3
(e) Show the numbering of anthracene molecule. Why do electrophilic aromatic substitutions occur most readily at 9-position in anthracene? 3

2. (a) Give the preparation of ketones from acid chlorides with organocopper compounds. Why Grignard reagents are not use in this reaction instead of organocopper compounds? 1×4
(b) How will you carry out the following conversions?
(i) Acetophenone into α-phenyl ethyl alcohol
(ii) Benzaldehyde into benzyl amine
(iii) Acetaldehyde into acetaldoxime
(iv) Benzoylchloride into benzophenone 2+2
(c) Explain the following:
(i) Aromatic aldehydes are less reactive than aliphatic aldehydes toward nucleophilic addition reaction.
(ii) The alpha hydrogen of a carbonyl compound is acidic. 3
(d) How are hemiacetals and acetals formed? Give mechanism. 2+2

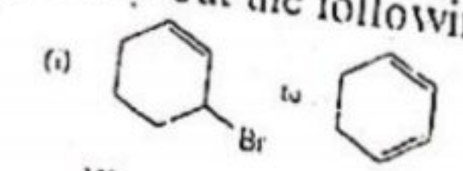
3. (a) How can you prepare carboxylic acids by
(i) carbonation of Grignard reagent and (ii) oxidation of alkyl benzene.
(b) What product would you expect to be formed when a carboxylic acid is treated with: 3
(i) $SOCl_2$, reflux; (ii) $LiAlH_4$ and (iii) C_2H_5OH, H^+.
(c) Why does an acyl chloride undergo nucleophilic attack more rapidly than does alkyl chloride. 3
(d) What is Hell-Volhard-Zelinsky reaction? How can you prepare α-hydroxy carboxylic acids and α,β-unsaturated carboxylic acids using this reaction? 4

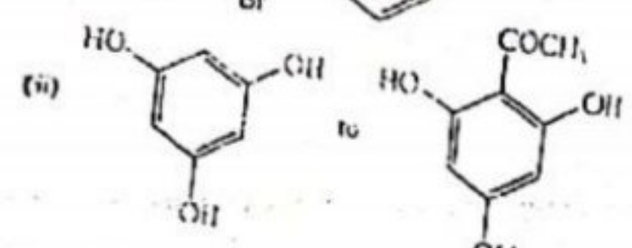
4. (a) Arrange the following compounds with increasing order of their basicity and justify your answer: Aniline, 4-Hydroxyaniline and 4-Nitroaniline. 3
(b) Synthesize the following compounds from aniline via benzene diazonium salt: 4.5
(i) Benzoic acid, (ii) Fluorobenzene and (iii) Phenol.
(c) Outline the synthesis of 1,2,3-tribromobenzene from 4-nitroaniline. 3
(d) Give a test to identify 1°, 2° and 3° nitroalkanes. 2.5
(e) Complete the following conversions: 1

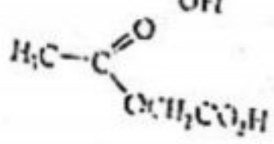
NH₂ → NHCOCH₃ → NHCOCH₃

5. (a) Give the reaction, mechanism and one synthetic application of the following: 3x
(i) Cannizzaro reaction, (ii) Michael reaction,
(iii) Diels-Alder reaction, (iv) Wittig reaction.
(b) How will you prove that Cannizzaro reaction involves a hydride transfer?

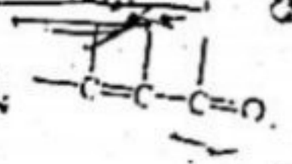
6. (a) Define the following terms with suitable example: 2
(i) Enantiomer, (ii) Diastereomer, (iii) Meso compound.
(b) Discuss on the elements of symmetry of organic molecules. 3
(c) What is racemic mixture? Give at least two methods to prepare a racemic mixture. 3
(d) Draw the orbital picture of 1,3-butadiene. 3
(e) Carry out the following conversions: 2

(i) 

(ii) 

(iii) $HOCH_2CO_2H$ to 

1x3

7. (a) How would you synthesize the following drugs:
(i) Sulphapyridine, (ii) Mepacrine, (iii) Paracetamol, (iv) Barbital, (v) Dulcin 2x5
(b) Discuss about the electrophilic and nucleophilic addition to the system,  4

8. (a) Give a method of synthesis for each of the following:
(i) Indole, (ii) Isoquinoline, (iii) Pyrimidine. 2x3
(b) Predict the products of reaction of indole with the following reagents:
(i) SO_3, pyridine, Δ; (ii) CH_3I, DMF, Δ; (iii) HCHO, $(CH_3)_2NH$; (iv) Sn, HCl and
(v) O_3, $HCONH_2$. 1x5
(d) Discuss the structure and chemistry of imidazole.

3.

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